

Exercise 1F

Understanding

- 1 -3 and 3 are opposites. Write down the opposites of these numbers.

a -6	b 10	c 38	d -46
e -32	f 88	g 673	h -349
- 2 Write the words 'add' or 'subtract' to suit each sentence.
 - a To add a negative number _____ its opposite.
 - b To subtract a negative number _____ its opposite.
- 3 Decide if the following statements are true or false.

a $5 + (-2) = 5 + 2$	b $3 + (-4) = 3 - 4$	c $-6 + (-4) = -6 - 4$
d $-1 + (-3) = 1 - 3$	e $8 - (-3) = 8 + 3$	f $2 - (-3) = 2 - 3$
g $-3 - (-1) = 3 + 1$	h $-7 - (-5) = -7 + 5$	i $-6 - (-3) = 6 + 3$

Example 11a, b

- 4 Evaluate the following.

a $6 + (-2)$	b $4 + (-1)$	c $7 + (-12)$	d $20 + (-5)$
e $2 + (-4)$	f $26 + (-40)$	g $-3 + (-6)$	h $-16 + (-5)$
i $-18 + (-20)$	j $-36 + (-50)$	k $-83 + (-22)$	l $-120 + (-139)$

Example 11c, d

- 5 Evaluate the following.

a $2 - (-3)$	b $4 - (-4)$	c $15 - (-6)$	d $24 - (-14)$
e $59 - (-13)$	f $147 - (-320)$	g $-5 - (-3)$	h $-8 - (-10)$
i $-13 - (-16)$	j $-10 - (-42)$	k $-88 - (-31)$	l $-125 - (-201)$
- 6 Write down the missing number.

a $4 + \underline{\hspace{1cm}} = 1$	b $6 + \underline{\hspace{1cm}} = 0$	c $-2 + \underline{\hspace{1cm}} = -1$
d $\underline{\hspace{1cm}} + (-8) = 2$	e $\underline{\hspace{1cm}} + (-5) = (-3)$	f $\underline{\hspace{1cm}} + (-3) = -17$
g $12 - \underline{\hspace{1cm}} = 14$	h $8 - \underline{\hspace{1cm}} = 12$	i $-1 - \underline{\hspace{1cm}} = 29$
j $\underline{\hspace{1cm}} - (-7) = 2$	k $\underline{\hspace{1cm}} - (-2) = (-4)$	l $\underline{\hspace{1cm}} - (-436) = 501$

Fluency

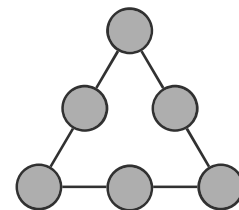
- 7 Place the integers from -3 to 2 in this magic triangle so that each side adds to the given number.

a -3	b 0
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- 8 A magic square has each row, column and main diagonal adding to the same magic sum. Complete these magic squares.

a	b
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		1
0	-2	-4

-12		
	-15	
	-11	-18



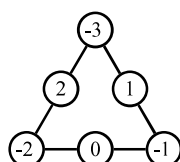
Problem-solving

- 8 a -1 b 8 c -7 d -1
 9 ground floor
 10 a - b -, + c -, -
 11 \$7
 12 -23°C
 13 a true b not always true c not always true
 d true e not always true f not always true
 14 a true b false c true
 15 0
 16 500, pair to give 500 pairs each with a total of 1.
 17 a a = 1, b = 4 b a = -7, b = 3
 c a = -5, b = 2 d a = -10, b = 2

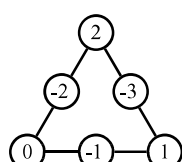
Exercise 1F

- 1 a 6 b -10 c -38 d 46
 e 32 f -88 g -673 h 349
 2 a subtract b add
 3 a false b true c true d false
 e true f false g false h true
 i false
 4 a 4 b 3 c -5 d 15
 e -2 f -14 g -9 h -21
 i -38 j -86 k -105 l -259
 5 a 5 b 8 c 21 d 38
 e 72 f 467 g -2 h 2
 i 3 j 32 k -57 l 76
 6 a -3 b -6 c 1 d 10
 e 2 f -14 g -2 h -4
 i -30 j -5 k -6 l 65

7 a



b



8 a

-1	-6	1
0	-2	-4
-5	2	-3

b

-12	-19	-14
-17	-15	-13
-16	-11	-18

9 a a = -3, b = -5

b a = -15, b = -9

10 -\$40

11 3 and -8

12 15 and -4

13 a Should be 5 + 2. b Left off negative on the 2.

14 a true b true c not always true d true

15 a no b yes c yes d no

16 a

x	-2	-1	0	1	2	3
y	7	6	5	4	3	2

b 18

c -45

17 a

x	-3	-2	-1	0	1	2	3
y	0	1	2	3	4	5	6

b 15

c -9

Exercise 1G

1 a

$\square$	$\triangle$	$\square \times \triangle$
3	5	15
2	5	10
1	5	5
0	5	0
-1	5	-5
-2	5	-10
-3	5	-15

b

$\square$	$\triangle$	$\square \times \triangle$
3	-5	-15
2	-5	-10
1	-5	-5
0	-5	0
-1	-5	5
-2	-5	10
-3	-5	15

2 a 15, 3 b -15, -3 c -15, -3 d 15, -3

3 a true b false c true

d true e false

4 a -20 b -54 c -40 d -99

e 6 f 105 g 400 h 300

i -152 j -123 k 54 l 765

5 a -5 b -2 c -4 d -30

e -2 f -26 g -45 h -36

i 3 j 3 k 9 l 6

6 a -3 b -5 c -7 d 6

e -3 f -72 g -252 h -5

i -30

7 a 1 b 14 c 160 d -29

e -120 f 20

8 a $\times, \div$ b $\times, \div$ c $\div, \times$ d $\div, \div$

e $\times, \div$ f $\times, \div$

9 0

10 -16

11 -8 and 3

12 8 and -2

13 a i 4 ii -27 iii -64 iv 25

b Yes, it will be a product of 2 numbers of the same sign.

c Yes, the product of 3 negative numbers will be negative.